A Comparison of Brain MRI Outcomes in Youth American Football versus Noncontact Sport Athletes

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ABSTRACT

ICHESCO, E., Y. LI, C.-H. SHIH, I. ICHESCO, A. ALMEIDA, E. VARANGIS, A. SCHREPF, C. KAPLAN, M. POPOVICH, S. J. PELTIER, R. E. HARRIS, M. T. LORINCZ, X. HE, and J. ECKNER. A Comparison of Brain MRI Outcomes in Youth American Football versus Noncontact Sport Athletes. *Med. Sci. Sports Exerc.*, Vol. 58, No. 2, pp. 261–270, 2026. **Objectives:** This study aimed to compare brain magnetic resonance imaging (MRI) outcomes between children who play American football versus noncontact sport controls testing the hypotheses that history (primary) and duration (secondary) of football participation would be associated with differences in cortical thickness, subcortical volume, resting-state functional connectivity, and white matter diffusivity. **Methods:** This secondary analysis of cross-sectional baseline data from the Adolescent Brain Cognitive Development Study compared brain MRI outcomes between 9- and 10-year-old children who play American football (n = 1194) versus noncontact sport controls (n = 807). Outcomes included 74 bilateral cortical thickness regions; 10 gray matter subcortical volumes, with *a priori* focus on the hippocampi; resting-state functional connectivity (169 network-network correlations and 247 network-region correlations across 13 resting-state functional networks and 19 regions); and 21 diffusion tensor measures. **Results:** Football participation was associated with global effects on cortical thickness (P = 0.017), network-to-network resting-state connectivity (P = 0.010), and fiber tract volume (false-discovery rate [FDR]-adjusted P = 0.015) in primary analysis, but the only significant post hoc finding after FDR correction was smaller cortical thickness adjacent to the left anterior transverse collateral sulcus in the football group (Cohen d = −0.258, FDR-adjusted P = 0.017). There were no significant duration of football play effects in secondary analyses (all P > 0.05). Targeted analysis of hippocampal volumes yielded no significant football or duration of play results (both P > 0.05), but suggested a potential trend of lower hippocampal volumes with increasing duration of play. **Conclusions:** At ages 9–10, participation in American football was associated with minimal differences across a large array of structural, functional, and diffusion tensor MRI outcomes. While the clinical implications of these cross-sectional results are unknown, they merit additional investigation and can contribute to the ongoing discussion surrounding contact sport participation in children. **Key Words:** ADOLESCENTS, BRAIN, DIFFUSION MRI, FOOTBALL, FUNCTIONAL MRI, STRUCTURAL MRI

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Youth contact sport participation is a highly discussed topic with significant public health implications. Beyond short-term concussion risk, there is concern that sport-related concussions (SRCs) and repetitive sport-associated head impacts may have long-term effects on brain health. This concern is magnified in youth athletes with developing brains (1,2). With over 670,000 children ages 6–12 and over 1.4 million adolescents ages 13–17 participating annually in American football alone (3), identifying potential effects of youth football participation is of critical societal importance.

The literature addressing the potential effects of early football exposure in former athletes is mixed. Early studies

by Stamm et al. in small samples of National Football League retirees raised concern with findings showing that those who began playing before age 12 had worse cognitive performance and altered corpus callosum microstructure compared with those who began playing later (4,5). These findings were not replicated by Solomon et al. (6) in a similar early investigation. However, Alosco et al. later found age of first exposure to football before age 12 was associated with worse cognitive, behavioral, mood, and functional outcomes in two additional studies (7,8). Additionally, Montenigro and colleagues found greater cumulative head impact index values associated with earlier age of exposure to be associated with greater cognitive and neurobehavioral deficits later in life (9). In contrast, Iverson et al. identified no differences between those who started playing football before versus after age 12 across multiple brain health outcomes (10,11). A recent study conducted by Terry and colleagues (12) including former professional football players similarly found that age of first exposure (before or after 12) was not associated with postcareer health outcomes. Further, Lempke et al. (13) found small but significant associations between earlier football exposure and multiple patient-reported outcomes, but not cognitive or mood diagnoses, in a larger National Football League retiree population. Finally, a small longitudinal study by Schaffert et al. (14) found no age of first exposure to football effects on baseline neuropsychological test results nor on change scores over 15–51 months of follow-up.

Research investigating early exposure to football and other contact sports in current athletes has more consistently failed to identify adverse effects. For example, Brett et al. (15) found no association between earlier football exposure and any behavioral, cognitive, psychological, or physical outcomes in a large high school and college football cohort. Similarly, Caccese et al. found no age of first-exposure effects on computerized neurocognitive test results nor on postconcussion recovery outcomes in two large college football cohorts (16,17). Similar results have been reported for other contact sports (16–19).

Collectively, existing research suggests possible population-level adverse effects of early contact sport exposure in former football athletes that do not present until later in adulthood, but earlier group-level effects in adolescent and young adult athletes have not been supported. However, a number of longitudinal studies in contact sport athletes from this age cohort do in fact support an association between nonconcussive head impacts sustained during football participation and short-term changes in measures of brain structure and function over the course of one or more seasons, most commonly involving neuroimaging outcomes (20–29). Yet, we are unaware of any large studies comparing magnetic resonance imaging (MRI)-based measures of brain structure and function between preadolescent youth contact sport athletes versus controls at a group level.

It is important to explore potential effects earlier in childhood when decisions can still be made regarding the degree and timing of contact sport participation. Therefore, the purpose of this study is to assess the effect of American football participation on MRI-based measures of brain structure and function in 9–10-year-old children. We focus specifically on American football because other contact sports do not typically allow body contact at this age. We leveraged the Adolescent Brain Cognitive Development (ABCD) study database to compare baseline MRI measures between children who participate in football versus noncontact sport controls. Given known beneficial effects of sport participation, exercise, and fitness on brain health (30–32), we designated noncontact sport athletes as controls to create groups that were as similar as possible except in terms of routine exposure to sport-associated head contact. We tested the hypotheses that history (primary) and duration (secondary) of football participation would be associated with cortical thickness, subcortical volume, resting-state functional connectivity, and white matter diffusivity. In addition to global analyses, we focused specific attention on hippocampal volumes given prior reports of lower hippocampal volumes in adult contact sport athletes and those with concussion (33–40).

METHODS

Data source and participants. The ABCD Study (https://abcdstudy.org/) recruited 11,875 children ages 9–10 years old across 21 United States research centers (41–46). Local Institutional Review Board approval was obtained at each site, and participants provided informed written consent/assent in accordance with the Declaration of Helsinki. Data are publicly available through the National Institute of Mental Health Data Archive. We used baseline ABCD neuroimaging and questionnaire data from Release 5.0 (NIMH Data Archive Digital Object Identifier 10.15154/8873-zj65; DOIs available at https://nda.nih.gov/study.html?id=2147 (44,46).

Participants with history of participation in either American football or any noncontact sport on the Sports Activities Involvement Questionnaire (SAIQ) were eligible (46). The SAIQ is a caregiver-report survey evaluating participation in sports, nonsport activities, and other hobbies. Participants with missing neuroimaging data were excluded, as were participants with low-quality resting-state functional MRI (rs-fMRI) data (<5-min data collection; voxel count in top or bottom 0.25 percentiles; average frames with displacement > 0.55 mm) (47). Participants with history of American football participation were classified into the football group, regardless of other sport history. Participants reporting only noncontact sport participation (Rice et al. (48) criteria) were classified as noncontact sport controls. This categorization was limited to participation in one or more of only

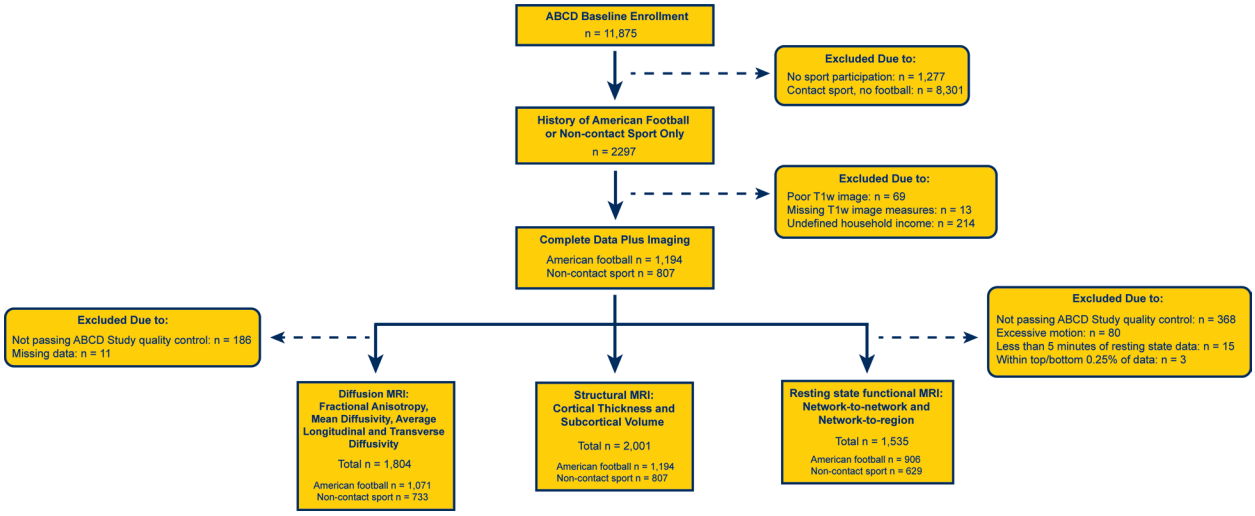

FIGURE 1—Inclusion/exclusion flow diagram of the participants in this study for each neuroimaging modality (i.e., cortical thickness/subcortical volume, functional resting-state connectivity, and diffusion).

the following sports: ballet/dance, climbing, swimming, tennis, track/running/cross country, and yoga/Tai Chi. Those with any other contact or limited contact sport participation history were excluded. Figure 1 presents an inclusion/exclusion flow diagram.

Neuroimaging outcomes. Participants completed the standard ABCD neuroimaging protocol, which underwent careful optimization and harmonization of MR scanners (44), including high-resolution T1-weighted structural MRI, rs-fMRI, and diffusion-weighted MRI, as previously described (44). Neuroimaging data were preprocessed and quality reviewed before public release as previously described (49). We utilized tabulated ABCD Neuroimaging Data Release 5.0.

Structural MRI outcomes included average cortical thickness values based on the Destrieux atlas parcellation yielding 74 bilateral surface parcels and five average bilateral subcortical gray matter volumes based on an automated, atlas-based volumetric segmentation procedure (49–51).

Resting-state fMRI outcomes included 169 network-network correlations and 247 network-region correlations across 13 resting-state functional networks and 19 regions of interests. Regions of interests were derived from the Destrieux atlas using a functionally defined parcellation based on resting-state functional connectivity patterns (52). Networks were derived from the Gordon parcellation (auditory, cingulo-opercular, default mode, dorsal attention, frontoparietal, retrosplenial, temporal, salience, sensorimotor hand, sensorimotor mouth, ventral attention, and visual).

Diffusion MRI outcomes included preprocessed measures of fractional anisotropy, mean diffusivity, average longitudinal and transverse diffusivity. Major white matter tracts were identified and labeled using AtlasTrack (53). Each diffusion metric was analyzed over five parcellations (tract, automatic segmentation parcel, subadjacent white matter, subadjacent gray matter, and gray matter-white matter contrast) with one additional whole-brain diffusivity measure, yielding a total of 21 global diffusion tensor analyses with between 30 and 68 individual elements.

Statistical analysis. Football participation history was classified as a binary categorical independent variable for primary analyses. Secondary analyses classified years of football participation as an additional ordinal independent variable ranging from zero (i.e., control group) to 4+ years. Children reporting 5 or more years of football play were combined into a single 4+ years category to maintain balanced group sizes, given diminishing numbers reporting more than 4 years of play.

All analyses accounted for age, biological sex, race, handedness, household income, ABCD study site, and traumatic brain injury (TBI) history as covariates. General demographic information was obtained from the parent/caregiver-reported demographics section of the battery designed by the ABCD Physical and Mental Health Assessment Workgroup, based on the PhenX Toolkit (46,54). Age was calculated as date of baseline study visit minus date of birth, in months. Biological sex was reported as assigned at birth. Race was based on parent/caregiver response to, "What race do you consider the child to be?" including all selected options (i.e., White and Black/African-American, Asian). Handedness was recorded as "right handed," "left handed," or "mixed handed" based on the Edinburgh Handedness Inventory Short Form (55). Household income was classified as one of eleven categories (<$5000, $5000–$11,999, $12,000–$15,999, $16,000–$24,999, $25,000–$34,999, $35,000–$49,999, $50,000–$74,999, $75,000–$99,999, $100,000–$199,999, ≥$200,000, and "don't know") based on parent/caregiver response to, "Which of these

categories best describes your total combined family income for the past 12 months?." TBI history was classified as positive or negative based on parent/caregiver response to the Ohio State University TBI Identification Method (56,57). Football-TBI interaction effects were also assessed. Models assessing subcortical gray matter volumes, rs-fMRI outcomes, and diffusion MRI outcomes accounted for total intracranial volume, mean motion, and average/maximum movement (i.e., framewise displacement), respectively, as additional covariates. Before analysis, box plots and quantile-quantile plots were reviewed to ensure a near-normal data distribution and rule out outliers. The number of participants varied slightly between models due to missing data. All statistical analyses were performed using R Version 3.5.2 (packages "stats" and "car").

Primary analyses fit separate multivariate general linear models evaluating global differences between the football and control groups across structural (2), rs-fMRI (2), and diffusion (23) outcomes. Wilk's likelihood ratio tested the null hypothesis of no difference between the football and control groups, accounting for potential confounders. Overall, football effects were deemed statistically significant if the null hypothesis was rejected at P <0.05 (unadjusted for structural and functional outcomes; Benjamini-Hochberg false-discovery rate [FDR] adjusted for diffusion outcomes). When significant global effects were present, post hoc t-tests identified specific regions, structures, and/or networks differing between groups. Adjusted P <0.05 after FDR correction was considered the threshold for significance and, when present, Cohen d effect sizes were calculated using pooled standard deviations. Given our *a priori* clinical interest in the hippocampus, hippocampal volumes were also specifically evaluated. Finally, Cook's distance and residual plots were evaluated for abnormal observations, heterogeneity, and nonlinear relationships.

Similar secondary analyses were performed including duration of football play as an additional variable. For significant post hoc results with FDR-adjusted P <0.05, additional linear models were fit with 1, 2, 3, and 4+ years of football participation as separate categorical variables to allow identification of potential nonlinear trends and to calculate effect size estimates with associated confidence intervals. Hippocampal volumes were again specifically evaluated.

Given the disproportionately high representation of males in the football group, additional male-only sensitivity analyses were performed. All models and analyses were otherwise identical except for the omission of biological sex as a covariate.

RESULTS

A total of 2001 children were eligible for inclusion: 1194 in the football group (447 with 1 year of football participation, 292 with 2 years, 217 with 3 years, 232 with ≥4 years; 6 with missing duration) and 807 noncontact sport controls (ballet/dance, n = 520; climbing, n = 27; swimming, n = 422; tennis, n = 73, track/running/cross country, n = 83; yoga/Tai Chi n = 41; some individuals reported participation in >1 noncontact sport). Demographics are presented in Table 1. More participants in the football group were males and identified as Black/African-American race compared with noncontact controls (Table 1). The distribution of study sites also differed between the groups (χ^2 [20, N = 2001] = 108.1, P < 0.001; median between-group difference by site <2%, maximum difference = 6.5%).

Structural MRI. In the primary cortical thickness analysis, there was a significant global football effect (smaller cortical thickness, P = 0.017; Supplemental Table 1a, Supplemental Digital Content, https://links.lww.com/MSS/D304) but no global effects of TBI (P = 0.081) nor football-TBI interaction effects (P = 0.303). Post hoc tests identified significantly smaller cortical thickness values adjacent to the left anterior transverse collateral sulcus (LATCS) in football players (Cohen d = −0.258, FDR-adjusted P = 0.019; Supplemental Table 2, Supplemental Digital Content, https://links.lww.com/MSS/D304). A similar significant global football effect was also present in the secondary cortical thickness analysis (P = 0.034; Supplemental Table 3a, Supplemental Digital Content, https://links.lww.com/MSS/D304), without global effects of duration of football play (P = 0.056), TBI (P = 0.084), nor football-TBI interaction effects (P = 0.298). Post hoc tests failed to identify significant football effects in any individual cortical thickness region after FDR correction (Supplemental Table 4, Supplemental Digital Content, https://links.lww.com/MSS/D304).

In the primary subcortical volume analysis, there were no global football (P = 0.508), TBI (P = 0.505), nor football-TBI interaction effects (P = 0.086; Supplemental Table 1a, Supplemental Digital Content, https://links.lww.com/MSS/D304). Focused hippocampal volume comparisons demonstrated no significant football effects on the right (P = 0.864) nor the left (P = 0.944). Similarly, there were no significant global football (P = 0.132), duration of football play (P = 0.083), TBI (P = 0.534), nor football-TBI interaction effects (P = 0.090) in the secondary analysis (Supplemental Table 3a, Supplemental Digital Content, https://links.lww.com/MSS/D304). Focused hippocampal volume comparisons demonstrated a trend toward decreasing right and left hippocampal volumes with increasing duration of football play, but this trend did not reach statistical significance after FDR correction (P = 0.062 for left and right; Table 2 and Fig. 2).

Resting-state functional MRI. In the primary analysis, there was a global football effect on network-to-network resting-state connectivity (P = 0.010); however, post hoc tests failed to identify significant football effects in any individual network pairs after FDR

TABLE 1. Demographics.

Variable	Football Group (n = 1194)	Noncontact Group (n = 807)	Effect Size	Significance (P Value)
Age	10.0 ± 0.6	9.9 ± 0.6	0.187	<0.001
Sex				
Male	1127 (94.4%)	183 (22.7%)	57.053	<0.001
Female	67 (5.6%)	624 (77.3%)	0.018	<0.001
Race				
White	584 (48.9%)	325 (40.3%)	1.420	<0.001
Black/African-American	315 (26.4%)	157 (19.5%)	1.483	<0.001
Asian/Pacific Islander	159 (13.3%)	222 (27.5%)	0.405	<0.001
AIAN/NHPI	8 (0.7%)	26 (3.2%)	0.206	<0.001
Other	128 (10.7%)	77 (9.5%)	1.138	0.437
Handedness				
Right	938 (78.5%)	649 (80.5%)	0.892	0.341
Left	89 (7.5%)	56 (6.9%)	1.079	0.728
Ambidextrious	167 (14.0%)	102 (12.6%)	1.123	0.424
TBI history				
Positive	66 (5.5%)	15 (1.9%)	3.063	<0.001
Negative	1128 (94.5%)	792 (98.1%)	0.327	<0.001
Annual household income				
<$5000	72 (6.0%)	43 (5.3%)	1.134	0.580
$5000–$11,999	55 (4.6%)	51 (6.3%)	0.696	0.083
$12,000–$15,999	39 (3.3%)	21 (2.6%)	1.138	0.716
$16,000–$24,999	66 (5.5%)	52 (6.4%)	0.848	0.433
$25,000–$34,999	86 (7.2%)	74 (9.2%)	0.782	0.157
$35,000–$49,999	93 (7.8%)	102 (12.6%)	0.587	<0.001
$50,000–$74,999	140 (11.7%)	99 (12.3%)	0.929	0.642
$75,000–$99,999	168 (14.1%)	102 (12.6%)	1.126	0.408
$100,000–$199,999	338 (28.3%)	193 (23.9%)	1.256	0.030
≥$200,000	137 (11.5%)	70 (8.7%)	1.450	0.018

Values represent mean (SD) for continuous variables and count (percentage) for categorical variables. Effect size estimates represent Cohen d (with pooled standard deviation) for continuous outcomes and odds ratios for categorical outcomes. P-values are from t-tests for continuous outcomes and chi-square tests are for categorical outcomes.
AIAN/NHPI, American Indian, Alaskan Native/Native Hawaiian, Pacific Islander.

correction (Supplemental Table 2, Supplemental Digital Content, https://links.lww.com/MSS/D304). There were no significant TBI (P = 0.829), nor football-TBI interaction effects (P = 0.459; Supplemental Table 1a, Supplemental Digital Content, https://links.lww.com/MSS/D304). A global football effect (P = 0.025) was again present in the secondary analysis, but there were no duration of play (P = 0.873), TBI (P = 0.832), nor football-TBI interaction effects (P = 0.454; Supplemental Table 3a, Supplemental Digital Content, https://links.lww.com/MSS/D304). Post hoc tests again failed to identify significant football effects in any individual network pairs after FDR correction (Supplemental Table 4, Supplemental Digital Content, https://links.lww.com/MSS/D304).

In the primary network-to-subcortical connectivity analysis, there were no significant global football (P = 0.676), TBI (P = 0.126), nor football-TBI interaction effects (P = 0.653; Supplemental Table 1a, Supplemental Digital Content, https://links.lww.com/MSS/D304). The secondary analysis similarly yielded no global football (P = 0.936), duration of football play (P = 0.509), TBI (P = 0.119), nor football-TBI interaction effects (P = 0.651; Supplemental Table 3a, Supplemental Digital Content, https://links.lww.com/MSS/D304).

Diffusion MRI. In the primary analysis, a significant global football effect was observed with decreased fiber tract volume (P = 0.015; Supplemental Table 1, Supplemental Digital Content, https://links.lww.com/MSS/D304); however, post hoc tests failed to identify

TABLE 2. Secondary results—hippocampal volume by years of football participation.

Football Years Played	Left Hippocampus Volume			Right Hippocampus Volume		
	Raw Effect	Effect size	P Value	Raw Effect	Effect Size	P Value
1 (n = 447)	20.827	0.051	0.376	14.114	0.033	0.577
2 (n = 292)	8.34	0.021	0.748	−0.656	−0.002	0.981
3 (n = 217)	−26.12	−0.064	0.360	−23.035	−0.054	0.453
4+ (n = 232)	−32.242	−0.079	0.250	−46.041	−0.107	0.127

Raw effects represent model-fit differences from the noncontact sport control group (in cubic mm). Effect sizes are represented as Cohen d, with a pooled standard deviation.

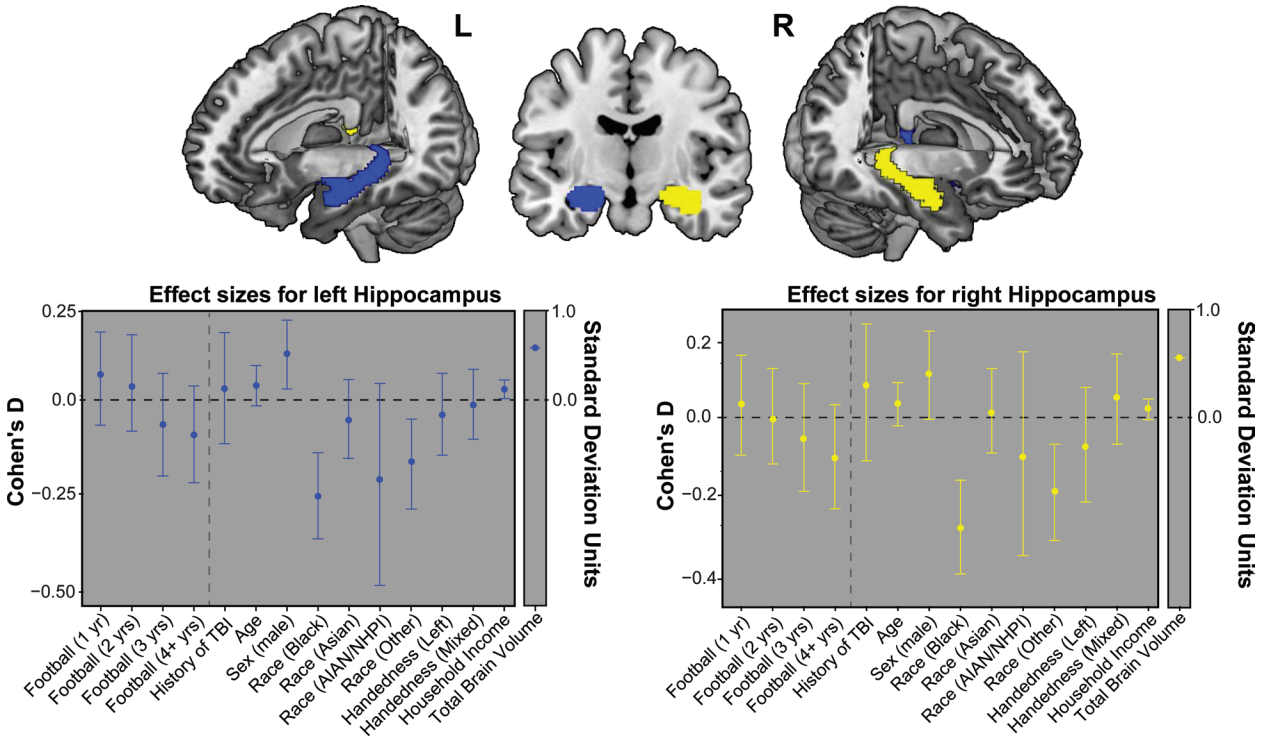

FIGURE 2—Hippocampal volumes. Bilateral subcortical hippocampal gyri regions in maize and blue are displayed. The bottom portion of the figure displays effect sizes of covariates in the model including football duration effects and additional covariates of no interest.

significant football effects in any individual fiber tract region after FDR correction (Supplemental Table 2, Supplemental Digital Content, https://links.lww.com/MSS/D304). No significant global football, TBI, nor football-TBI interaction effects were present for any of the remaining 20 diffusion MRI metrics in the primary analysis (Supplemental Table 1b, Supplemental Digital Content, https://links.lww.com/MSS/D304), nor were significant football, duration of football play, TBI, nor football-TBI interaction effects present across any of the 21 diffusion MRI metrics in the secondary analysis (Supplemental Table 3b, Supplemental Digital Content, https://links.lww.com/MSS/D304).

Sensitivity analysis. Excluding female participants had no meaningful effect on the results. Differences from the original full sample results were minimal and included the following: the primary analysis global football effect on network-to-network connectivity was not significant (P = 0.069); a secondary analysis global football duration effect was present for cortical thickness (P = 0.039), without significant post hoc findings in any individual region; global football-TBI interaction effects were present in primary and secondary analyses for subcortical volume (both P = 0.002), with post hoc results reaching significance for the left amygdala (both P = 0.009) driven by an unreliably small sample size of only four control participants with history of TBI (Supplemental Fig. 1, Supplemental Digital Content, https://links.lww.com/MSS/D304).

DISCUSSION

To our knowledge, this is the largest neuroimaging study to investigate relationships between football participation and brain structure and function in children. Our use of ABCD data allowed us to evaluate these relationships in a large "naturalistic" national sample. Our primary analyses across numerous structural, functional, and diffusion MRI outcomes identified only a single cortical thickness region differing between the football and noncontact control groups. Further, our secondary analyses failed to identify any significant dose-effect relationships associated with years of youth football participation. Our targeted analysis of hippocampal volumes found neither a significant overall football effect nor a significant duration of play effect, although a nonsignificant trend of lower hippocampal volumes with greater duration of football participation was observed.

Smaller LATCS cortical thickness values in the football group were the only statistically significant primary analysis findings. The LATCS lies anteriorly in the medial temporal lobe between the parahippocampal and fusiform gyri and is known to have substantial anatomical variability (58–60). Given its proximity to the limbic system, it is thought to play a role in memory function (58–60), but there is little published research explicitly describing clinical correlates. One study in a cohort of 125 retired athletes over age 50 with positive concussion histories identified greater LATCS gray matter volumes

(left and right combined) compared with matched controls (40), while a different study in nonathletes found lower LATCS cortical thickness values in a small cohort of heroin-dependent adults compared with controls (61). While statistically significant, the associated effect size was small, and ultimately, this finding is of unclear clinical significance. Future analyses of the ABCD dataset should investigate potential correlates between these LATCS findings and memory outcomes.

While our hippocampal volume findings were nonsignificant, an apparent trend of lower hippocampal volumes with increasing duration of football play was observed. Smaller hippocampal volumes have previously been reported in adult collision sport (e.g., football and rugby) athletes. Singh et al. (33) compared hippocampal volumes in collegiate football athletes with and without history of TBI to matched controls and found both significant football effects and significant concussion effects. Further, years of football play were significantly inversely correlated with left hippocampal volumes in the football athletes. Meier et al. (38) found inverse associations between bilateral hippocampal volumes and the number of self-reported concussions in collegiate football athletes without duration of play effects. Additionally, 3 publications by researchers at Stanford University reported smaller right hippocampal volumes in football versus volleyball athletes and decreasing CA1 volumes over time in those football athletes without concussion (34–36). Additionally, Arciniega et al. (39) found lower hippocampal volumes in a cohort of 170 male former professional and collegiate American football athletes compared with controls.

However, lower hippocampal volumes in contact sport athletes are not a universal finding. More similar to our own results, Brett et al. (62) found no overall difference in hippocampal volumes between collegiate contact sport athletes and matched controls, and actually found larger hippocampal volumes in those contact sport athletes with history of concussion compared with those without. Bobholz et al. (63) subsequently reported neither between-group differences in hippocampal volumes between concussed collegiate athletes versus contact- and noncontact sport controls nor hippocampal volume changes over time in the concussed cohort. While others have previously compared hippocampal volumes between contact and noncontact sport athletes, a major difference of our current investigation is that our younger study population is still undergoing critical aspects of brain development. While we did not find significant football effects on hippocampal volumes, our observed trend of lower volumes with longer duration of football participation merits follow-up.

The presence of significant primary global football effects for network-to-network resting state connectivity and fiber tract volume not localizing to any individual network or region is of unclear clinical significance. It is possible that small differences may be present across multiple networks or regions that are of insufficient magnitude to achieve significance on their own but are collectively large enough to drive the significant global effects observed. Indeed, each individual network-network pair and fiber track region approaching statistical significance (i.e., with uncorrected $P < 0.05$) has an effect size of D <0.2 (Supplemental Table 2, Supplemental Digital Content, https://links.lww.com/MSS/D304) confirming small magnitude effects. Ultimately, whether these nonlocalizing significant global effects are attributable to cumulative small effect sizes or an alternative underlying explanation, their presence merits additional follow-up.

Despite this study's strengths, several limitations must be considered. First, causality cannot be inferred from this secondary analysis of observational, cross-sectional ABCD data. Additional prospective research that is also able to account for potential within-individual changes over the course of a season is necessary. While our statistical analysis accounted for several covariates with potential influence on brain structure and function (64–66), other potential moderating effects may exist. The potential for sex-based moderation is of particular concern given the degree of imbalance in males and females between groups, but our males-only sensitivity analysis speaks against biological sex as a potential confounder. Despite the lower number of control participants in this male-only analysis, we elected not to include additional male nonathletes because the potential benefit of more balanced groups would be offset by the potential lack of general sport and exercise/fitness effects in the additional nonathlete male controls. There are also limitations intrinsic to the self-reported ABCD data. The SAIQ does not discriminate between tackle and flag football participation, so it is possible that some children in our American football cohort may participate in flag football, which is classified as a limited contact sport (48). This could potentially minimize the observed effects present in our football group. Further, the Ohio State University TBI Identification Method uses more stringent criteria for identifying TBI than are typically applied for SRC, so underestimation of SRC history is possible. Inaccurately self-reported data is also possible. Up to 9 years of football participation was recorded for some children suggesting initiation of play around one year of age. Our creation of a "4+" years of play subgroup mitigates concern for over-reporting of duration of play, but inaccurate self-reporting of other data elements is possible. Another potential limitation specific to the ABCD neuroimaging data collected on different MRI scanners across 21 sites is the possibility of site- and/or scanner-specific differences. While the ABCD study has gone to great lengths to standardize MRI results across sites, differences may exist. Finally, this investigation

focused only on MRI-based measures of brain structure and function. Future research should broaden its scope to include additional cognitive, mood, and behavioral outcomes.

CONCLUSIONS

While the minimal presence of significant structural, functional, and diffusion MRI-based findings associated with American football participation offers some reassurance with respect to potential risks of early contact sport exposure in children, the fact that any significant findings were observed in this young cohort also raises some potential concern. The smaller LATCS cortical thickness values in the football group, the nonlocalizing global football effects for network-to-network resting-state connectivity and fiber tract volume, and the potential dose-effect trend between duration of football play and hippocampal volumes all merit additional longitudinal follow-up. These current results can be incorporated into the ongoing discussion within the sports medicine community regarding timing of contact sport participation in children.

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